Solutions



One of the common mixtures you will use in Science is the solution. A **solution** is a type of mixture containing a **solute dissolved** in a **solvent**. A **solute** is a solid which will dissolve in a solvent, and the solid is said to be **soluble**. A **solvent** is a substance which will dissolve a solute. For example, when sugar dissolves in water, sugar is the solute and water is the solvent. A substance which **dissolves** in another is said to **mix evenly into it**. A solid which will not dissolve in a solvent is said to be **insoluble**. Two liquids which mix evenly in with each other are said to be **miscible**. Two miscible liquids are petrol and kerosene. Two liquids which will not dissolve in each other are called **immiscible**. Petrol and water are immiscible.

1. Complete the following for the words, **soluble and miscible**.

Vocabulary

What does it mean?

Find a word that is:

Similar

Opposite

Word: SOLUBLE

Use it in a sentence

Draw a picture of it

(4 marks)

Vocabulary

What does it mean?

Find a word that is:

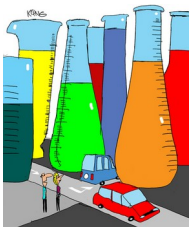
Similar

Opposite

Word: MISCIBLE

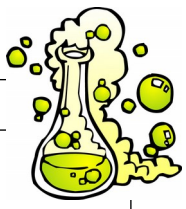
Use it in a sentence

Draw a picture of it



(4 marks)

Complete the following table using the paragraph on the previous page:



Term (Word)	Definition
Solution	
Solute	
Solvent	
Insoluble	
Dissolve	
Immiscible	

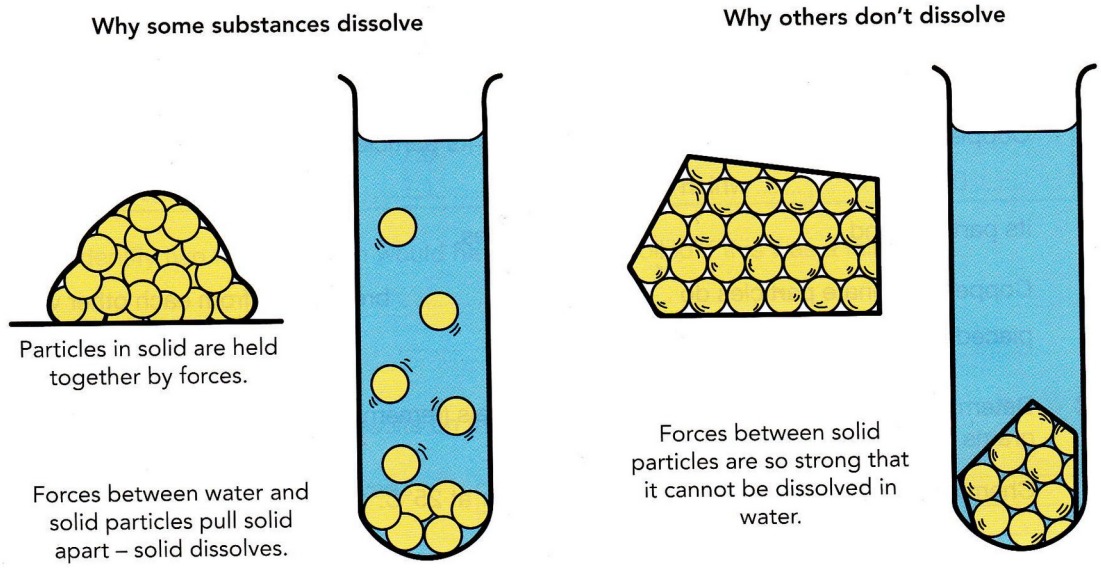
(6 marks)

SOLUBILITY

Substances are **soluble** if they dissolve in a solvent. Substances are **soluble** if the forces holding their particles together are **weaker** than the forces between their particles and those of the solvent.

Substances which do not dissolve in a particular liquid are said to be **insoluble** in that liquid. Substances are **insoluble** if the forces holding their particles together are **stronger** than the forces between their particles and those of the solvent.

The **solubility** of a substance is a measure of how much of it dissolves in 100mL of water at 25°C.



Complete these sentences to explain why some substances are soluble and others are insoluble. Use the following words:

chemical, particles, water, forces, chemical, strength, particles

1. When a chemical is put in water, three sets of forces act between particles.

   There will be forces of attraction between the water _____ and the particles of the _____.

   There will be forces of attraction between water _____ and other _____ particles.

   There will be forces of attraction between the particles within the _____ itself.

   Whether or not the chemical dissolves depends on the _____ of each of these three _____ .

   (6 marks)

water, chemical, attraction

2. If the force of _____ between _____ particles and the particles of the _____ is greater than the other two forces, then the chemical will dissolve.

   (3 marks)

other, tightly, dissolve, will, each, not

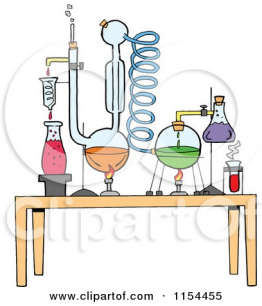
3. If the particles of the chemical are held together so _____ that the water particles cannot break them away from_____ _____ then the chemical _____ _____ _____.

   (6 marks)

particles, soluble

4. Sodium chloride (common salt) is _____ in water because its _____ break away from each other when it is put into water.

   (2 marks)



GRAPHING

5. The table shows the results of an experiment done to examine how the solubility of three different chemicals changed with temperature. Use them to answer the questions.

Temperature (°C)	Amount of chemical dissolving in 100mL water		
	Chemical A	Chemical B	Chemical C
10	3.1	8.2	0.4
20	4.0	7.3	0.8
30	4.7	6.7	1.2
50	5.8	5.5	2.6
60	6.3	5.0	3.6
70	6.7	4.5	4.8
80	7.0	4.2	6.6
100	7.1	3.5	12.0

(i) Graph the three sets of results on the one graph. (line graph)

(5 marks)

(ii) Identify two things that would have been controlled if this was a fair test.

_____

_____

(2 marks)

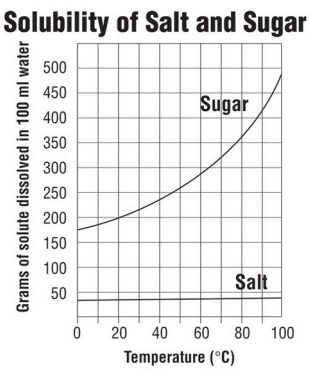
(iii) Which of the chemicals is most soluble at:

a) 35°C _____
b) 65°C _____

(2 marks)

(iv) If you have 1 litre of water. At what temperature could you dissolve:

a) 58g of A _____
b) 42g of B _____
c) 26g of C _____



(3 marks)

(v) What mass of A would dissolve in 1 L of water at 35oC? _____

(vi) What mass of B would dissolve in 1 L of water at 45oC? _____

(vii) What mass of C would dissolve in 1 L of water at 55oC? _____

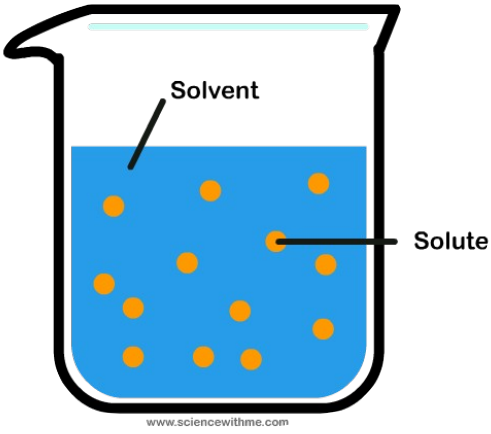
(3 marks)

(viii) At what temperature is B twice as soluble as A? _____

(ix) At what temperature is C half as soluble as B? _____

(x) At what temperature is B three times more soluble than C? _____

(3 marks)



Solution